

## Diffusion of niobium in $\alpha$ -iron

Naoko Oono\*, Hiroyuki Nitta\* and Yoshiaki Iijima

Department of Materials Science, Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan

Diffusion coefficient of  $^{95}\text{Nb}$  in  $\alpha$ -iron has been determined in the temperature range between 823 and 1163 K by use of a serial sputter-microsectioning technique. The diffusion coefficient of niobium is about two times as large as the self-diffusion coefficient in  $\alpha$ -iron. The temperature dependence of the diffusion coefficient,  $D$ , in the whole temperature range of  $\alpha$ -iron across the Curie temperature ( $T_C = 1043$  K) can be expressed by  $D = (1.40_{-1.05}^{+4.17}) \times 10^{-1} \exp \left[ -\frac{(299.7 \pm 12.7 \text{ kJmol}^{-1})(1 + (0.061 \pm 0.007)s^2)}{RT} \right] \text{ m}^2\text{s}^{-1}$  where  $s$  is the ratio of the spontaneous magnetization at  $T$  K to that at 0 K. The factor 0.061 in the equation is smaller than 0.156 for the self-diffusion, showing that the magnetic effect on the diffusion of niobium in  $\alpha$ -iron is smaller than that on the self-diffusion. The activation energy  $299.7 \text{ kJmol}^{-1}$  for niobium diffusion in the paramagnetic iron is much higher than  $250.6 \text{ kJmol}^{-1}$  for the self-diffusion. Atomic size effect is predominant in the activation energies for the diffusion of transition metal solutes in the paramagnetic  $\alpha$ -iron.

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### 1. Introduction

Diffusion data on iron-base alloys are very important in evaluating the behavior for high temperature creep, oxidation, grain boundary corrosion, *etc.* of the materials in operation. Addition of niobium improves generally the heat resistance and creep strength of steels.<sup>1)</sup> Niobium is also used for the stabilization of carbon in stainless steels and for grain size control and precipitation hardening in high strength low alloy steels.<sup>2)</sup> A knowledge of the diffusion behavior of niobium in iron and steels is therefore important for a full understanding of these processes.

In the past, diffusion of niobium in iron has been studied by Geise and Herzig<sup>3)</sup> and by Herzig *et al.*<sup>4)</sup> using radioactive tracer  $^{95}\text{Nb}$  and the serial sectioning technique with precision grinding and microtome. However, the diffusion coefficient of niobium in  $\alpha$ -iron obtained by these authors<sup>3,4)</sup> is about one order of magnitude larger than the self-diffusion coefficient.<sup>5,6)</sup> It seems that their values are anomalously high in comparison with the diffusion coefficients of transition metal solutes in  $\alpha$ -iron.<sup>7)</sup> In the studies on self-diffusion in  $\alpha$ -iron,<sup>5,6)</sup> it has been noted that the diffusion coefficient obtained by use of conventional mechanical sectioning methods is larger than that obtained by the sputter-microsectioning technique. In the experiments with mechanical sectioning methods, the average diffusion distance,  $2\sqrt{Dt}$  ( $D$ : diffusion coefficient,  $t$ : diffusion time), is usually in the range 30–100  $\mu\text{m}$  which is greater than normal inter-dislocation spacing in  $\alpha$ -iron. As a result, the measured diffusion coefficients in  $\alpha$ -iron obtained by the conventional mechanical sectioning techniques are affected by the enhancement effect due to the diffusion along the dislocation networks. In particular, the influence of short circuit paths on diffusion is more serious at low temperatures; *i.e.* the  $\gamma$ - $\alpha$  transformation temperature of iron corresponds to  $0.65T_m$  ( $T_m$ : melting temperature). In the previous experiments of the diffusion of niobium in  $\alpha$ -iron,<sup>3,4)</sup> the average diffusion distance in the paramagnetic  $\alpha$ -iron was in the range from 33 to 90  $\mu\text{m}$ .

Therefore, the large diffusion coefficient of niobium in  $\alpha$ -iron<sup>3,4)</sup> was probably caused by the enhanced diffusion along short-circuit paths.

The activation energy for diffusion of niobium in iron is also interesting. A linear relationship between the atomic radius and the activation energy for the diffusion of transition metal solutes in the paramagnetic  $\alpha$ -iron has been observed.<sup>8)</sup> The atomic radius of niobium is about 15% larger than that of iron,<sup>9)</sup> which is, in the view of Hume Rothery's rule, nearly the upper limit to form a solid solution with iron, and thereby the solid solubility of niobium in  $\alpha$ -iron and  $\gamma$ -iron is very low.<sup>10)</sup>

Below the Curie temperature, the Arrhenius plot of diffusion coefficients of solute as well as solvent atom in iron of the ferromagnetic state deviates downwards from the Arrhenius relationship extrapolated from the paramagnetic state.<sup>11)</sup> The temperature dependence of the diffusion coefficient in the whole temperature range across the Curie temperature has been well described by<sup>12)</sup>

$$D = D_0^P \exp[-Q^P(1 + \alpha s^2)/RT]. \quad (1)$$

Here  $D_0^P$  and  $Q^P$  are respectively the preexponential factor and the activation energy for the diffusion in the paramagnetic state. The value of  $s$ , the ratio of the spontaneous magnetization at  $T$  K to that at 0 K, has been experimentally determined by Potter<sup>13)</sup> and by Crangle and Goodman.<sup>14)</sup> The constant  $\alpha$  expressing the extent of the influence of the magnetic transformation on diffusion, is given by

$$\alpha = \frac{\alpha_f + \alpha_m}{Q^P}, \quad (2)$$

where  $\alpha_f$  and  $\alpha_m$  are the increments of formation energy and migration energy, respectively, of a vacancy by the magnetic transformation.

The purpose of this work is to study the diffusion behavior of niobium in high purity  $\alpha$ -iron in a wide range of temperature by use of a sputter-microsectioning technique and radioactive tracer  $^{95}\text{Nb}$ .

\*Graduate Student, Tohoku University

Table 1 Chemical analysis of iron rod (mass ppm).

| C | N | O  | P | S  | Cr | Ni | Si |
|---|---|----|---|----|----|----|----|
| 1 | 2 | 37 | 2 | <1 | <1 | 7  | 3  |

## 2. Experimental Procedure

High purity iron rod was supplied by Mitsubishi Heavy Industries, Ltd. The rod was prepared as follows; high purity electrolytic iron, "Miron HP" made by Toho Zinc Co., was induction-melted in a cold-copper crucible under  $2 \times 10^{-4}$  Pa after evacuation of the chamber to a base pressure of  $10^{-7}$  Pa. The ingot was hot-forged and machined in the form of a rod 12 mm in diameter. The chemical analysis of the rod is shown in Table 1. The rod was cut to make the disc specimen 1 mm in thickness. The flat face of the specimen was ground on abrasive papers and electropolished in an aqueous solution containing 77 vol% acetic acid and 18 vol% perchloric acid at 273 K. To induce grain growth and to reduce the contents of carbon, nitrogen and oxygen, the specimen was annealed in the  $\delta$ -iron phase at 1773 K for 24 h in a stream of hydrogen gas purified by permeation through a palladium tube at 700 K. By this treatment the gaseous contents would be less than 1 mass ppm.<sup>15)</sup> The resultant grain size was about 3–5 mm. After the grain growth, the specimen was ground on abrasive papers and polished with fine alumina paste and finally electropolished under the same conditions as mentioned above. To obtain a stress-free surface, the specimen was further annealed at 1123 K for 7.2 ks in the stream of purified hydrogen gas.

The radioisotope  $^{95}\text{Nb}$  ( $\gamma$ -rays, 0.766 MeV; half-life, 35 h) was supplied in the form of niobium chloride in 6.0M HCl by NEN Life Science Products, Inc. USA. The radionuclidic purity of the isotope was 99.5%. Taking care to avoid oxidation of the specimen, the radioisotope was electroplated on the surface of the specimen with  $14 \text{ mA cm}^{-2}$  for 0.5–2.0 min. The specimens were then sealed in quartz tubes under  $10^{-4}$  Pa and diffused at a temperature in the range from 823 to 1163 K for 1.8 to 550.8 ks in furnaces controlled to within  $\pm 0.5$  K. Some specimens were diffused in a stream of the purified hydrogen gas. To measure the penetration profiles of the diffusing species into the specimen, three types of the serial sputter-microsectioning apparatus were used; the ion beam type with the sputtering rate of 7–10 nm/min and the radio-frequency type with the sputtering rate of 30–60 nm/min were used to measure short penetration profiles and the magnetron type with the sputtering rate of 150–300 nm/min was used to measure long penetration profiles. The details of the methods are described elsewhere.<sup>16,17)</sup> For each specimen, 15–35 successive sections were sputtered. A constant fraction of the sputtered-off material was collected on an aluminum foil. The intensity of the radioactivity of the isotope in each section was measured by a well-type Tl-activated NaI detector with a 1024 channels pulse height analyzer.

## 3. Results

For one-dimensional volume diffusion of a tracer from an infinitesimally thin surface layer into a sufficiently long rod, the solution of Fick's second law is given by

$$I(X, t) \propto C(X, t) = [M/\sqrt{\pi Dt}] \exp(-X^2/4Dt) \quad (3)$$

where  $I(X, t)$  and  $C(X, t)$  are respectively the intensity of the radioactivity and the concentration of the tracer at a distance  $X$  from the original surface after a diffusion time  $t$ .  $D$  is the volume diffusion coefficient of the tracer and  $M$  is the total amount of tracer deposited on the surface before the diffusion.

Figures 1 and 2 show the plots of  $\ln I(X, t)$  vs.  $X^2$  for the diffusion of  $^{95}\text{Nb}$  in  $\alpha$ -iron at higher and lower temperatures, respectively. In accordance with eq. (3), all the plots in Figs. 1 and 2 are linear, indicating that the concentration profiles are governed by the volume diffusion. As seen in Fig. 1, very near the surface, a few points shown by open circles deviate apparently from a linear line for the volume diffusion because of oxidation of the tracer on the surface. The diffusion coefficients calculated from the slope of the plot are listed in Table 2, and their Arrhenius plots are shown in Fig. 3 along with the diffusion coefficients obtained by Geise and Herzig<sup>3)</sup> and by Herzig *et al.*<sup>4)</sup> The temperature dependence of the self-diffusion coefficient in  $\alpha$ -iron is also drawn in Fig. 3. The self-diffusion coefficients obtained by Iijima *et al.*,<sup>5)</sup> Hettich *et al.*<sup>18)</sup> and Lübbhusen and Mehrer<sup>6)</sup> are in good agreement with each other and hence, individual experimental points are not shown in Fig. 3 to avoid the cluttering of data points. It is seen that the diffusion coefficient of niobium determined by the present work is about two times as large as the self-

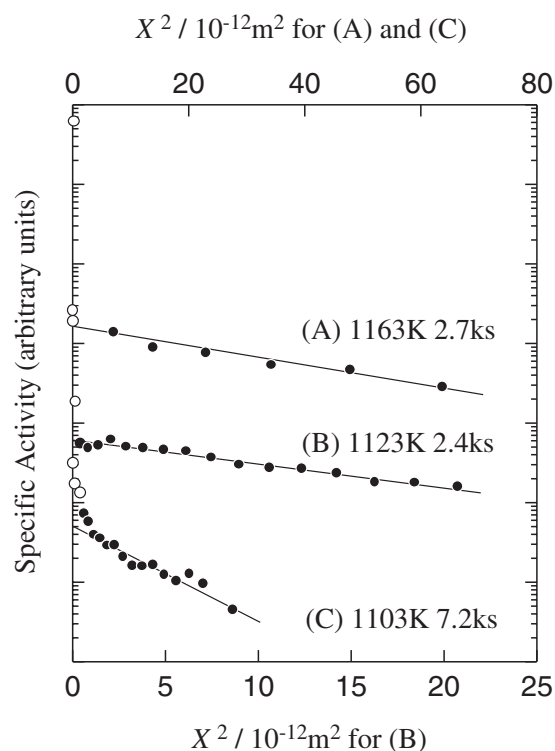


Fig. 1 Examples of penetration profiles for diffusion of  $^{95}\text{Nb}$  in the paramagnetic  $\alpha$ -iron.

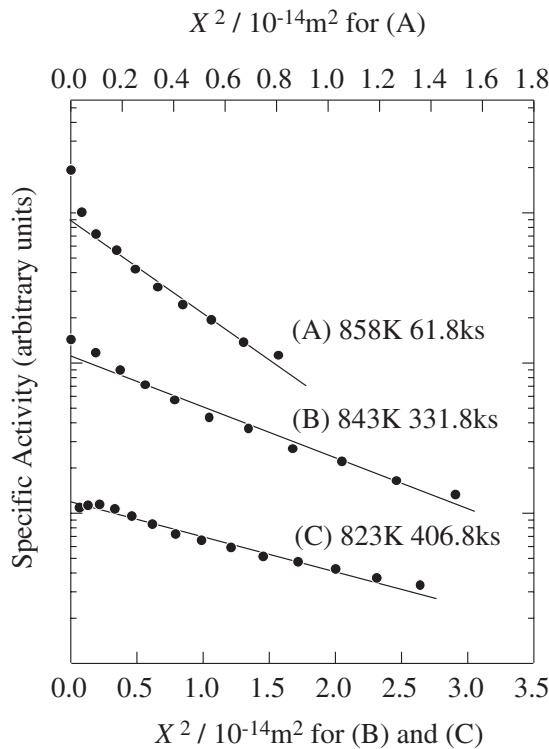


Fig. 2 Examples of penetration profiles for diffusion of <sup>95</sup>Nb in the ferromagnetic  $\alpha$ -iron.

Table 2 Diffusion coefficient of niobium in  $\alpha$ -iron.

| Temperature<br>$T/K$ | Diffusion time<br>$t/ks$ | Diffusion coefficient<br>$D/m^2s^{-1}$ | Technique |
|----------------------|--------------------------|----------------------------------------|-----------|
| 1163                 | 2.7                      | $4.04 \times 10^{-15}$                 | MS        |
| 1148                 | 7.2                      | $3.61 \times 10^{-15}$                 | MS        |
| 1133                 | 1.8                      | $2.07 \times 10^{-15}$                 | MS        |
| 1123                 | 1.8                      | $1.50 \times 10^{-15}$                 | MS        |
| 1123                 | 2.4                      | $1.56 \times 10^{-15}$                 | RF        |
| 1103                 | 2.7                      | $9.78 \times 10^{-16}$                 | RF        |
| 1103                 | 7.2                      | $1.00 \times 10^{-15}$                 | MS        |
| 1083                 | 4.2                      | $5.42 \times 10^{-16}$                 | MS        |
| 1073                 | 2.7                      | $3.86 \times 10^{-16}$                 | MS        |
| 1053                 | 62.4                     | $1.53 \times 10^{-16}$                 | MS        |
| 1033                 | 58.8                     | $5.48 \times 10^{-17}$                 | MS        |
| 993                  | 10.8                     | $1.58 \times 10^{-17}$                 | IBS       |
| 983                  | 86.4                     | $1.43 \times 10^{-17}$                 | RF        |
| 973                  | 201.6                    | $7.29 \times 10^{-18}$                 | RF        |
| 923                  | 14.4                     | $5.04 \times 10^{-19}$                 | RF        |
| 888                  | 59.4                     | $8.58 \times 10^{-20}$                 | IBS       |
| 873                  | 64.8                     | $4.05 \times 10^{-20}$                 | IBS       |
| 858                  | 61.8                     | $1.76 \times 10^{-20}$                 | IBS       |
| 843                  | 331.8                    | $8.87 \times 10^{-21}$                 | IBS       |
| 823                  | 406.8                    | $3.37 \times 10^{-21}$                 | IBS       |

MS: magnetron sputter-microsectioning.  
RF: radio frequency sputter-microsectioning.  
IBS: ion-beam sputter-microsectioning.

diffusion coefficient, while that by Geise and Herzig<sup>3)</sup> and by Herzig *et al.*<sup>4)</sup> is one to two orders of magnitude larger than the self-diffusion coefficient. In the temperature range between the Curie temperature ( $T_c = 1043$  K) and the  $\alpha$ - $\gamma$

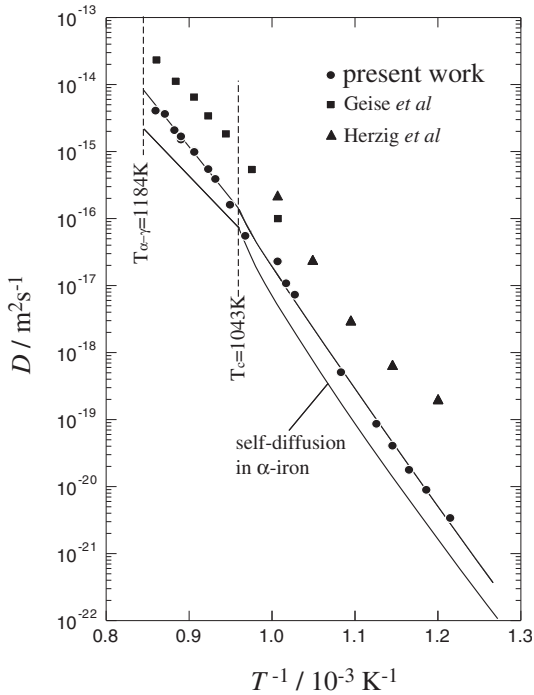


Fig. 3 Arrhenius plot of diffusion coefficients of niobium in  $\alpha$ -iron in comparison with those by other authors. Arrhenius line for self-diffusion in  $\alpha$ -iron is also shown.

transformation temperature ( $T_{\alpha-\gamma} = 1184$  K) the diffusion coefficient obtained by the present work shows a linear Arrhenius behavior with the pre-exponential factor,  $D_0^P$ ,  $(1.40^{+4.17}_{-1.05}) \times 10^{-1} m^2 s^{-1}$  and the activation energy,  $Q^P$ ,  $(299.7 \pm 12.7)$  kJmol<sup>-1</sup>. In  $\alpha$ -iron of the ferromagnetic state the diffusion coefficient of niobium is smaller than that extrapolated from the paramagnetic state, as is the case for the self-diffusion.<sup>5,6,18)</sup>

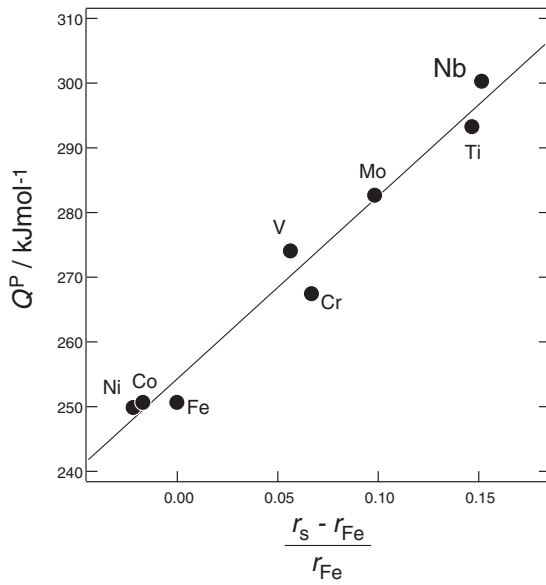
#### 4. Discussion

As seen in Figs. 1 and 2, the penetration profiles for diffusion of niobium in  $\alpha$ -iron show the Gaussian behavior in accordance with the lattice diffusion. However, it seems that the diffusion depth with the linear relation is somewhat short in comparison with those for chromium<sup>19)</sup> and cobalt<sup>20)</sup> in  $\alpha$ -iron. These Gaussian profiles are usually followed by tails due to the diffusion along short-circuit paths such as dislocations and grain boundaries.<sup>21)</sup> As mentioned earlier, niobium has a large atomic radius<sup>9)</sup> and small solubility in  $\alpha$ -iron<sup>10)</sup> in comparison with other transition metal solutes. Hence, niobium has an easy tendency to segregate at dislocations or grain boundaries in  $\alpha$ -iron. This characteristic of niobium is probably concerned with the short Gaussian profiles.

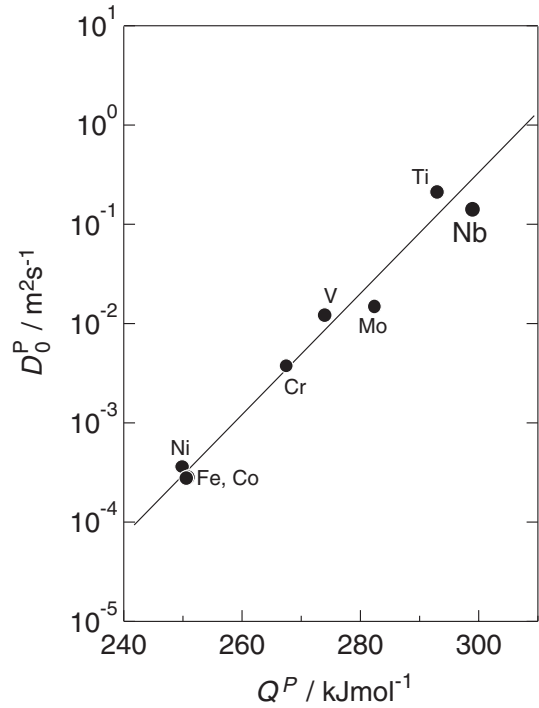
The diffusion coefficient of niobium is about twice the self-diffusion coefficient, as shown in Fig. 3. The enhanced diffusion can be explained by the larger atomic radius of niobium. The distortion of the iron lattice due to the oversized solute atom results in attractive interactions between solute atoms and vacancies. This Gibbs free enthalpy of binding between a vacancy and a solute atom leads to a decrease in the vacancy formation enthalpy and

Table 3 Preexponential factor, activation energy and constant  $\alpha$  for self-diffusion and diffusion of niobium, molybdenum, cobalt and chromium in  $\alpha$ -iron.

| Diffusant  | $D_0^p / \text{m}^2 \text{s}^{-1}$    | $Q^p / \text{kJmol}^{-1}$ | $Q_{\text{cal.}}^p / \text{kJmol}^{-1}$ | $\alpha$          | $Q^f = Q^p(1 + \alpha) / \text{kJmol}^{-1}$ |
|------------|---------------------------------------|---------------------------|-----------------------------------------|-------------------|---------------------------------------------|
| Iron       | $2.76_{-1.04}^{+1.42} \times 10^{-4}$ | $250.6 \pm 3.8$           | $250.5 \pm 0.4$                         | $0.156 \pm 0.003$ | $289.7 \pm 5.1$                             |
| Niobium    | $1.40_{-1.05}^{+4.17} \times 10^{-1}$ | $299.7 \pm 12.7$          | $299.0 \pm 8.8$                         | $0.061 \pm 0.007$ | $318.0 \pm 15.6$                            |
| Molybdenum | $1.48_{-0.75}^{+1.51} \times 10^{-2}$ | $282.6 \pm 6.4$           | $283.1 \pm 0.6$                         | $0.074 \pm 0.005$ | $303.5 \pm 7.5$                             |
| Cobalt     | $2.76 \times 10^{-4}$                 | 251                       | -                                       | 0.23              | 309                                         |
| Chromium   | $3.73_{-1.40}^{+2.24} \times 10^{-4}$ | $267.1 \pm 4.4$           | $267.5 \pm 0.5$                         | $0.133 \pm 0.005$ | $303.0 \pm 6.3$                             |

Fig. 4 Plot of  $Q^p$  vs.  $(r_s - r_{\text{Fe}})/r_{\text{Fe}}$ .

thus to an increase in diffusion.<sup>22)</sup> Table 3 summarizes the Arrhenius parameters for the self-diffusion<sup>5)</sup> and the solute diffusion of molybdenum,<sup>8)</sup> chromium,<sup>19)</sup> cobalt<sup>20)</sup> and niobium obtained by our research group using the radioactive tracer method with serial sputter-microsectioning techniques. The activation energies for diffusion of niobium and molybdenum in the paramagnetic  $\alpha$ -iron are larger than that for self-diffusion. Figure 4 shows a plot of the activation energy  $Q^p$  for self-diffusion,<sup>5)</sup> solute diffusion of transition metals of molybdenum,<sup>8)</sup> chromium,<sup>19)</sup> cobalt,<sup>20)</sup> titanium,<sup>22)</sup> nickel,<sup>23)</sup> vanadium<sup>24)</sup> and niobium obtained by the present work in the paramagnetic  $\alpha$ -iron against  $(r_s - r_{\text{Fe}})/r_{\text{Fe}}$ , where  $r_s$  and  $r_{\text{Fe}}$  are the radii of the solute atom for the coordination number 8 and of the iron matrix lattice.<sup>9)</sup> A linear relationship is observed in Fig. 4. Therefore, atomic size effect is predominant in the activation energies for the diffusion of transition metal solutes in the paramagnetic  $\alpha$ -iron. As described above, the solute atom with larger atomic radius induces higher concentration of vacancies around the solute. On the other hand, such large solute atom requires high enthalpy for migration in the matrix. Consequently, as shown in Fig. 4, the solute with large atomic size requires large activation energy for diffusion in the paramagnetic  $\alpha$ -iron. Furthermore, a linear relationship between the logarithm of the preexponential factor,  $D_0^p$ , and the activation energy,  $Q^p$ , for diffusion in the paramagnetic  $\alpha$ -iron is observed in Fig. 5. The relation is represented by

Fig. 5 Plot of  $D_0^p$  against  $Q^p$  for diffusion of transition elements in  $\alpha$ -iron.

$$\ln D_0^p (\text{m}^2 \text{s}^{-1}) = 1.37 \times 10^{-4} Q^p (\text{Jmol}^{-1}) - 42.4 \quad (4)$$

The magnitudes of both constants in the above equation are consistent with those for the diffusion by the vacancy mechanism.<sup>25,26)</sup>

To evaluate the magnetic contribution to the activation energy for diffusion of niobium in the ferromagnetic iron, eq. (1) is rewritten as follows

$$T \ln \frac{D(T)}{D_0^p} = -\frac{Q^p}{R} - \left( \frac{\alpha Q^p}{R} \right) s^2. \quad (5)$$

Substituting the values of  $D(T)$  and  $D_0^p$  obtained by the present work and the empirical value of  $s(T)$  for pure iron obtained by Potter<sup>13)</sup> and Crangle and Goodman<sup>14)</sup> into the eq. (5),  $T \ln[D(T)/D_0^p]$  was calculated as a function of  $s^2$  and plotted in Fig. 6. A linear relationship between  $T \ln[D(T)/D_0^p]$  and  $s^2$  in Fig. 6 shows that in iron of the ferromagnetic state the temperature dependence of diffusion coefficient of niobium obeys eq. (1). From the slope and intercept of the straight line in Fig. 6, the values of  $\alpha$  and  $Q^p$  were determined and listed in Table 3. These are compared with the values for self-diffusion<sup>5)</sup> and for diffusion of chromium<sup>19)</sup> and cobalt<sup>20)</sup> and molybdenum<sup>8)</sup> in  $\alpha$ -iron. Here,

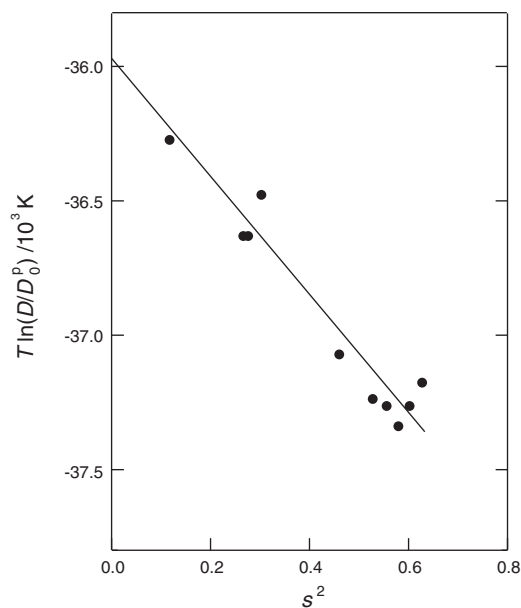


Fig. 6 Plot of  $T \ln[D(T)/D_0^p]$  vs  $s^2$  for diffusion of niobium in the ferromagnetic  $\alpha$ -iron.

the value of  $Q^p$  calculated in this way is denoted by  $Q_{\text{cal}}^p$ . It is noted that the value of  $Q^p$  determined directly from the Arrhenius plot of  $D(T)$  in the paramagnetic state is in good agreement with  $Q_{\text{cal}}^p$ , as shown in Table 3. Using the values of  $\alpha$  ( $= 0.061$ ),  $D_0^p$  ( $= 1.40 \times 10^{-1} \text{ m}^2 \text{ s}^{-1}$ ) and  $Q^p$  ( $= 299.7 \text{ kJ mol}^{-1}$ ) obtained by the present work,  $D(T)$  is calculated, and its Arrhenius plot is drawn as a solid line in Fig. 3. As shown in Table 3, the value of  $\alpha$  ( $= 0.061$ ) for the diffusion of niobium in  $\alpha$ -iron is about 60% smaller than 0.156 for the self-diffusion<sup>5)</sup>, indicating that the effect of magnetic transformation in iron on the solute diffusion of niobium is smaller than that on the self-diffusion. Using the value of  $Q^p$  ( $= 299.7 \text{ kJ mol}^{-1}$ ) for the diffusion of niobium in the paramagnetic iron, the activation energy  $Q^f$  ( $= (1 + \alpha)Q_p$ ) in the ferromagnetic iron can be estimated to be  $318.0 \text{ kJ mol}^{-1}$ . Thus, the increment of the activation energy  $\alpha Q^p$  by the magnetic transformation is  $18.3 \text{ kJ mol}^{-1}$  for the diffusion of niobium, while it is  $39.1 \text{ kJ mol}^{-1}$  for the self-diffusion. Thus, in the paramagnetic iron the activation energy for the diffusion of niobium is  $49.1 \text{ kJ mol}^{-1}$  larger than that for self-diffusion, whereas in the ferromagnetic region, this difference is reduced to  $28.3 \text{ kJ mol}^{-1}$ . The value of  $\alpha s^2$  can be taken as a measure of the degree of the influence of the magnetic transformation on diffusion. Putting  $D(T)$ ,  $D_0^p$  and  $Q^p$  into eq. (1), the value of  $\alpha s^2$  for diffusion of niobium is calculated as a function of temperature and shown in Fig. 7. The values of  $\alpha s^2$  for self-diffusion<sup>5)</sup> and for the diffusion of molybdenum,<sup>8)</sup> chromium<sup>19)</sup> and cobalt<sup>20)</sup> are also shown in Fig. 7. In the case of self-diffusion and diffusion of molybdenum, chromium and niobium in  $\alpha$ -iron,  $\alpha s^2$  decreases gradually with increase of temperature and becomes zero at and above the Curie temperature of pure iron. On the other hand, for the diffusion of cobalt in  $\alpha$ -iron,  $\alpha s^2$  is larger than that for the self-diffusion in the whole temperature range and becomes zero at and above 1140 K which is 97 K higher than the Curie temperature of pure iron.

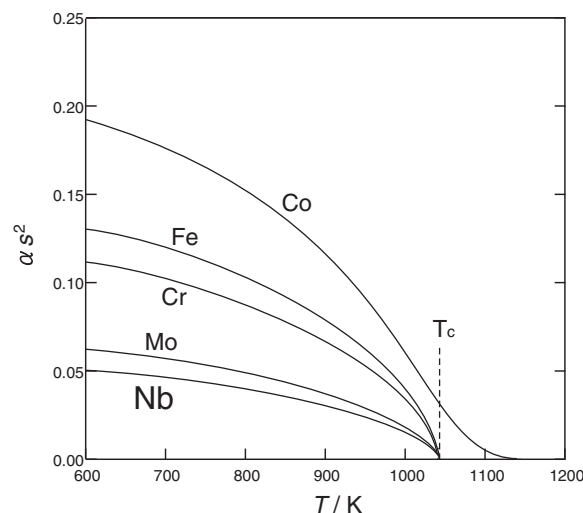


Fig. 7 Temperature dependence of  $\alpha s^2$  for diffusion of niobium in comparison with those for self-diffusion and diffusion of chromium, cobalt and molybdenum in  $\alpha$ -iron.

This means that a local magnetic field around a solute cobalt atom is stronger than that of pure iron matrix and it remains so even above the Curie temperature. However, in the case of chromium, molybdenum and niobium solute atoms, the magnetic field is weaker than that of the pure iron matrix and becomes zero at the Curie temperature. The experimentally determined values of the change in magnetization  $\Delta M$  of  $\alpha$ -iron on diluting with 3d, 4d and 5d elements are compiled by Pepperhoff and Acet.<sup>27)</sup> They are  $+1.0 \mu_B/\text{atom}$  for cobalt,  $-2.6 \mu_B/\text{atom}$  for chromium,  $-2.3 \mu_B/\text{atom}$  for molybdenum and  $-2.75 \mu_B/\text{atom}$  for niobium. The value  $\Delta M$  consists of contribution arising from the influence of the impurity atoms on the magnetic moment of the surrounding iron atoms and from the contribution of the local impurity moment. The latter is the difference between the iron atom moment and the local impurity moment. Thus,  $\Delta M$  is the sum of contributions from all neighbors extending out to several shells. According to theoretical calculations of  $\Delta M$  for transition-metal impurities in iron by Drittler *et al.*,<sup>28)</sup> the values of  $\Delta M$  are  $+0.79 \mu_B$  for cobalt,  $-3.29 \mu_B$  for chromium,  $-3.31 \mu_B$  for molybdenum and  $-3.68 \mu_B$  for niobium, although the local moments of impurities in iron are  $+1.65 \mu_B$  for cobalt,  $-1.29 \mu_B$  for chromium,  $-0.52 \mu_B$  for molybdenum and  $-0.46 \mu_B$  for niobium. Drittler *et al.*<sup>28)</sup> have calculated the change of the total moment summed over the impurity and five shells of iron atoms. At present, it is difficult to connect the value of  $\Delta M$  to the values of  $\alpha_f$  and  $\alpha_m$  in eq. (2). However, it can be concluded that the value of  $\Delta M$  is related qualitatively to the order of magnitudes of  $\alpha s^2$  for these solutes in Fig. 7.

## 5. Conclusion

Using the serial sputter-microsectioning techniques with radioactive tracer  $^{95}\text{Nb}$ , the diffusion coefficient of niobium in  $\alpha$ -iron has been determined in the temperature range from 823 to 1163 K. The diffusion coefficient of niobium is about twice the self-diffusion coefficient in  $\alpha$ -iron. The temperature



dependence of the diffusion coefficient of niobium is influenced by the magnetic transformation in a way similar to that for the self-diffusion. The increase in activation energy due to the magnetic transformation for solute diffusion of niobium is found to be  $18.3 \text{ kJmol}^{-1}$  which is smaller than the corresponding value of  $39.1 \text{ kJmol}^{-1}$  for the self-diffusion of iron.

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